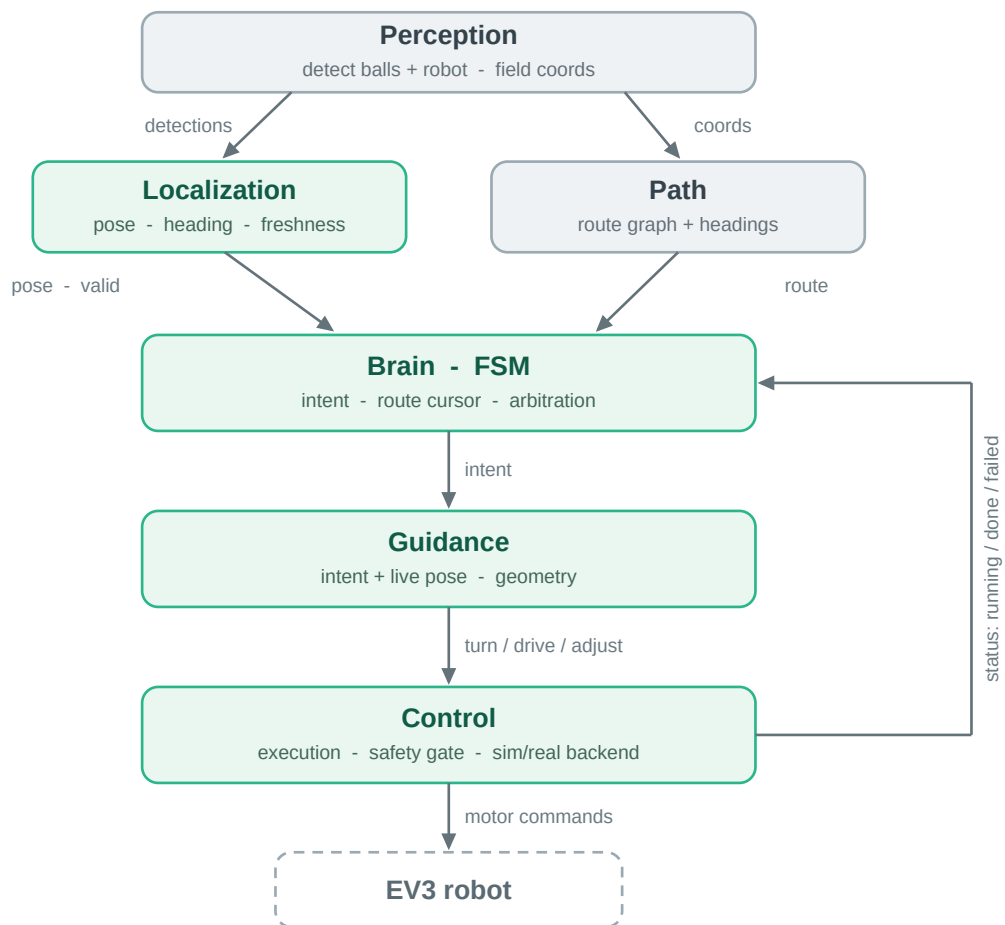


Proposed Layer Structure

Brainstorm synthesis · v0.1 · 15 Jun 2026. Scope: reinventing the robot movement layer. Perception and Path are kept as fixed inputs and are not in scope for this rework.

Each layer is named for the single thing it **owns**, so there is never a question of where a responsibility lives. The split maps loosely onto the classic robotics decomposition: where am I (Localization), what move reaches the goal (Guidance), make the motors do it (Control).



Layer stack. Teal = new layers to build; grey = kept (existing); dashed = hardware. Solid arrows are the downward data/command flow; the right-hand arrow is the upward status channel that closes the loop.

The layers

Perception
kept

Detects balls and the robot and outputs their positions in field centimetres. In: camera. Out: detections.

Localization new · thin	Turns detections into a clean robot pose - heading estimation, smoothing, and a freshness / validity flag so everything downstream knows whether to trust the pose this frame. In: detections. Out: RobotPose + valid.
Path kept	Converts the coordinate field into a route graph with headings, plus target selection and pickup / unload structures. In: coordinates. Out: route graph.
Brain (FSM) new	Decides intent - follow the route, turn to the next waypoint, pick up, unload, recover. Owns the route-progress cursor and arbitration between modes. Emits intent, not motor commands. In: pose + route. Out: intent.
Guidance new	Computes the geometry: given an intent and the live pose, produces turn / drive / adjust commands. In: intent + pose. Out: geometric commands.
Control new	Executes commands on the EV3 - speed profiles, 'close enough' thresholds, motor translation. Hosts the safety / validity gate as its final stage and a real-vs-simulated backend behind one command API. In: turn / drive / adjust. Out: motor commands, plus status upward.

Contracts that cross the boundaries

- **Status is a loop, not a pipe.** Every layer below the Brain returns running / done / failed upward; the Brain cannot choose the next intent without knowing the current one finished.
- **Perception fans out.** It feeds Localization and Path in parallel, and the Brain pulls from both - they are not in series.
- **Units and frames are a written contract.** cm vs mm, radians vs degrees, where heading zero points, which way positive rotation goes. Pin it once: every new boundary is a chance for a sign flip.
- **Safety / validity gate.** Lives as Control's final stage - stale pose, off-field, or an un-executable command stops here rather than being special-cased across the FSM.
- **Simulation seam.** Control hides a real-vs-simulated backend behind one command API, so the whole stack runs and is testable without the EV3.

The open hinge

One decision is deliberately deferred, and it lands on the Guidance / Control seam: do commands **block** until done, or **tick** with status feedback every frame? With ticked commands the six-layer split holds as drawn and the Brain stays live - it can see while the robot moves. With blocking commands, Guidance and Control tend to collapse into one 'do this move, report when done' layer, and *adjust* stops being a separate command. Treat Guidance as its own layer as provisional until this is settled: the boundary is real, but whether it is a wall or a hairline depends on this one call.

Naming notes

Names follow what each layer owns. The one to reconsider is **Guidance** - it is borrowed from aerospace and precise, but *Motion* or *Maneuver* carry the same meaning if the team reads them faster. Everything else holds.